

HP Smart Storage Administrator Scripting Cookbook

Abstract

This document provides basic and advanced scripting terminology and scenarios for HP Smart Storage Administrator users. The primary audience is the system administrator with a good working knowledge of storage hardware and the configuration of logical drives and arrays.



Part Number: 742647-002
January 2014
Edition: 2

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Overview

About HP SSA

HP SSA is the main tool for configuring arrays on HP Smart Array controllers. It exists in three interface formats: the HP SSA GUI, the HP SSA CLI, and HP SSA Scripting. All formats provide support for configuration tasks. Some of the advanced tasks are available in only one format.

The diagnostic features in HP SSA are also available in the standalone software HP Smart Storage Administrator Diagnostics Utility CLI.

Starting with HP SSA and ProLiant Gen8 servers and server blades, HP SSA is accessible both offline and online:

- Accessing HP SSA in the offline environment
Using one of multiple methods, you can run HP SSA before launching the host operating system. In offline mode, users can configure or maintain detected and supported ProLiant devices, such as optional Smart Array controllers and integrated Smart Array controllers. Some HP SSA features are only available in the offline environment, such as setting the boot controller and boot volume.
- Accessing HP SSA in the online environment
This method requires an administrator to download the HP SSA executables and install them. You can run HP SSA online after launching the host operating system.

For more information, see the *HP Smart Storage Administrator User Guide* on the HP website (<http://www.hp.com/go/smartstorage/docs>).

How to use this guide

This guide builds upon the instructions and explanation of the scripting functions for the HP Array Configuration Utility. The scenarios provided here are scripting-based solutions to the most common array configuration needs. Each scenario outlines the task, the hardware configuration, the scripting solution, and a summary that explains the use of certain options and their values.

The guide covers basic scripting scenarios that use general commands familiar to ACU or HP SSA users. For advanced users, the guide also contains advanced commands only available with the latest versions of HP SSA. Advanced scripting scenarios show how to combine basic and advanced commands for users' configuration needs.

Before using these scenarios, become familiar with the HP SSA scripting capabilities, specifically:

- Capturing a configuration
- Using an input script
- Creating HP SSA script files
- Understanding script file options

For information about these capabilities, and general HP SSA usage, see the *Configuring Arrays on HP Smart Array Controllers Reference Guide* on the Documentation CD that ships with the controller or on the HP

website

(<http://h18004.www1.hp.com/products/servers/proliantstorage/software-management/acumatrix/index.html>).

Changes to the Method mode Auto option

When creating a script, you must select options for the Action mode and the Method mode. In earlier versions of ACU Scripting, the `Auto` option was the default value for the Method mode. `Auto` is no longer the preferred option in HP SSA, and the Method mode no longer has a default option.

Beginning with ACU v8.28.13, HP changed the underlying library of rules that the software uses to optimize a configuration when employing the `Auto` option. The new library of rules is limited and is not backward compatible with existing configurations.

To provide greater flexibility and control during configuration, HP developed the advanced scripting commands outlined in this guide. HP recommends that users create a custom script with these commands to resolve the original configuration problems addressed by the `Auto` option.

Sample custom input script

The sample script in this section gives all possible values for each option.

- If an **option** is shown in bold type, you must enter a value for that option when writing your own script.
- Default options are not universal across controllers and firmware.

You can use this script as a template for your own script.

Action = `Configure|Reconfigure`

Method = `Custom`

Controller = `All|Slot [N][:N] | WWN [N] | First | SerialNumber [N] | IOCabinet [N],IOBay [N],IOChassis [N],Slot [N],Cabinet [N],Cell [N]`

`ClearConfigurationWithDataLoss = Yes|No |Forced`

`LicenseKey = XXXXX-XXXXX-XXXXX-XXXXX-XXXXX`

`DeleteLicenseKey = XXXXX-XXXXX-XXXXX-XXXXX-XXXXX | * ; COMMENT: * is a wild card that enables you to delete all license keys on the specified controller.`

`ReadCache = 0|10|20|25|30|40|50|60|70|75|80|90|100`

`WriteCache = 0|10|20|25|30|40|50|60|70|75|80|90|100`

`RebuildPriority = Low|Medium|Mediumhigh|High`

`ExpandPriority = Low|Medium|High`

`SurfaceScanDelay = N`

`SurfaceScanDelayExtended = N`

`SurfaceScanMode = Idle|High|Disabled`

`DriveWriteCache = Enable|Disable`

`MNPDelay = 0..60`

`IRPEnable = Enable|Disable`

`DOPEnable = Enable|Disable`

`ElevatorSortEnable = Enable|Disable`

`QueueDepth = 2|4|8|16|32|Automatic`

```

DriveWriteCache = Enable | Disable
PreferredPathMode = Auto|Manual

; Array Options
; There can be multiple array specifications in the file
Array = A|B|C|D|E|F|G|...Z|AA|AB|AC... | NEXT
OnlineSpareMode = Dedicated | AutoReplace
OnlineSpare = Port:ID,... | Box:Bay,... | Port:Box:Bay | None | N
Drive = Port:ID... | Box:Bay... | Port:Box:Bay,... | N | *

; Caching Array Options
; There can be only one Caching Array specifications in the file
CachingArray = A|B|C|D|E|F|...Z|AA|AB|AC...
Drive = Port:ID,... | Box:Bay,... | Port:Box:Bay,...

; Logical Drive Options
; There can be multiple logical drive specifications in the file
; The maximum strip size depends on the number of drives in an array and the
size of the controller cache
LogicalDrive = 1|2|3... max Volumes | NEXT
Repeat = 0... max Volumes
RAID = 0|1|10|5|6|ADG|50|60
Size = N|Max|MAXMBR
Sizeblocks = N
ShrinkSize = N
NumberOfParityGroups = N
NPG (same as NumberOfParityGroups)
Sectors = 32|63
StripSize = 8|16|32|64|128|256|512|1024
ArrayAccelerator = Enable|Disable

; Caching Logical Drive Options
CachingLogicalDrive = 1|2|3... max Volumes/2
RAID = 0|1 ;FW and Controller dependent
Size = N

```

Sample custom capture script

Running SSA Scripting in capture mode captures the configuration of all array controllers on the server in a capture file. The capture file can be used in input mode with minor edits to replicate the configuration to other array controllers.

Use of the "-internal" or "-external" command line switches effectively filters the controllers captured. For example, if the "-external" switch is used, then any controllers internal to the host computer are not captured in the output file.

Example:

```
; Date captured: Wed Jan 15 15:33:14 2014
; Version: 1:60:0:5

Action= Configure
Method= Custom

; Controller Specifications
; Controller HP Smart Array P800, Firmware Version 5.20
Controller= Slot 9
ReadCache= 25
WriteCache= 75
RebuildPriority= Medium
ExpandPriority= Medium
SurfaceScanDelay= 3
DriveWriteCache= Disabled
MNPDelay= 60
IRPEnable= Disabled
DPOEnable= Disabled
ElevatorSortEnable= Enabled
QueueDepth= Automatic
; Unassigned Drives
; 2E:2:1 (300 GB), 2E:2:8 (146 GB), 2E:2:9 (146 GB), 2E:2:10 (146 GB), 2E:2:11
(300 GB), 2E:2:12 (300 GB), 2E:2:13 (146 GB), 2E:2:14 (146 GB), 2E:2:15 (146 GB),
2E:2:16 (146 GB), 2E:2:17 (146 GB), 2E:2:18 (146 GB)

; Array Specifications
Array= A
; Array Drive Type is SAS
; 2E:1:1 (146 GB)
Drive= 2E:1:1
OnlineSpare= No

; Logical Drive Specifications
LogicalDrive= 1
RAID= 0
Size= 139979
Sectors= 32
StripeSize= 128
ArrayAccelerator= Enabled
```

Basic scripting scenarios

Scenario 1: Configure a two-disk array

Task

Create a simple two-disk array with a 100 GB RAID 1 logical volume.

Hardware configuration

- Storage system
- An unconfigured HP Smart Array controller in slot 1
If the controller contains an existing configuration, the script fails unless you add the -reset parameter to the command line to clear the existing configuration.
- Two available physical drives at the following location: 11:4:8 and 11:4:7

Script

```
Action=Configure
Method=Custom

Controller=Slot1

Array=A
Drive=11:4:8, 11:4:7

LogicalDrive=1
RAID=0
Size=100000
```

Summary

The script combines the two physical drives into a single 100-GB logical drive.

Scenario 2: Reconfigure to modify logical drive size

Task

Reconfigure an existing 100-GB logical volume to modify its size to 200 GB.

Hardware configuration

- Storage system
- An HP Smart Array controller in slot 1
- An existing logical drive with a capacity equal to or larger than 200 GB

Script

```
Action=ReConfigure
```



```
Method=Custom

Controller=Slot1

Array=A
LogicalDrive=1
RAID=0
Size=200000
```

Summary

The script modifies the size of the logical drive only, because multiple transformations of a logical drive cannot be performed at the same time. For example, HP SSA must complete the size expansion before beginning another transformation, such as a RAID level change.

To modify existing arrays and volumes, the HP SSA GUI may be a more intuitive tool for some users. The GUI provides graphical feedback that guides the user and displays only available options.

Scenario 3: Reconfigure to modify RAID level

Task

Reconfigure an existing RAID 0 logical volume to be a RAID 1 logical volume.

The selected RAID level is supported only if the minimum number of physical drives are available.

Hardware configuration

- Storage system
- An HP Smart Array controller in slot 1
- An existing logical drive

Script

```
Action=ReConfigure
Method=Custom

Controller=Slot1

Array=A
LogicalDrive=1
RAID=1
```

Summary

The script modifies the RAID level of the logical drive only, because multiple transformations of a logical drive cannot be performed at the same time. For example, HP SSA must complete the RAID level change before beginning another transformation, such as a size expansion.

To modify existing arrays and volumes, the HP SSA GUI may be a more intuitive tool for some users. The GUI provides graphical feedback that guides the user and displays only available options. For example, the HP SSA GUI only displays optional RAID levels based on the minimum number of required physical drives. For more information, see the *Configuring Arrays on HP Smart Array Controllers Reference Guide*.

Scenario 4: Configure by disk type and count

Task

Create an array by specifying the type and count of disks for HP SSA to use. This example use two SAS drives. This method does not require you to specify physical disks by location.

Hardware configuration

- Storage system
- An HP Smart Array controller in slot 1
- Two available SAS physical disks

This script fails if two unused SAS drives are not available on the controller.

Script

```
Action=Configure  
Method=Custom
```

```
Controller=Slot1
```

```
Array=A  
Drive=2  
DriveType=SAS  
LogicalDrive=1  
RAID=0
```

Summary

This script does not specify a size, but uses the `MAX` default value to consume the fully available capacity of the physical drives.

HP SSA does not support arrays with mixed drive types. An array must comprise drives of the same type, for example all SAS or all SATA drives. Different arrays can have different drive types.

Scenario 5: Configure two arrays, one with a spare

Task

Create two arrays with one online spare on a controller. One array uses two physical drives in a RAID 1 logical volume. The second array uses the remaining physical disks to create 20 identically sized RAID 5 logical volumes.

Hardware configuration

- Storage system
- An unconfigured HP Smart Array controller in slot 1

If the controller contains an existing configuration, the script fails unless you add the `-reset` parameter to the command line to clear the existing configuration.

- Numerous available physical drives at unspecified locations

Script

```
Action=Configure
```

```
Method=Custom

Controller=Slot1

Array=A
Drive=2
OnlineSpare=1
LogicalDrive=1
RAID=1
Size=MAX

Array=B
Drive=*
LogicalDrive=NEXT
Repeat=20
RAID=5
Size=MAX
```

Summary

For both arrays, this script uses the `MAX` value to consume the fully available capacity of the physical drives.

For Array B, the `Repeat` option specifies the number of times to repeat the configuration. Instead of using the number 2, the script uses the `NEXT` value to specify the ID of the second logical drive. When using the `Repeat` option, you must also use the `NEXT` value for the `LogicalDrive` option.

This script will fail if the controller does not have the required number of disks connected.

Advanced commands

Controller=ALL

You can use `Controller=All` instead of the `Slot=N` command.

Advantages

- You do not need to determine where controllers are installed.
- You can target all controllers at once.

Examples

Change read cache to 100% on all controllers (only configured controllers, those with volumes, accept this command):

```
Controller=ALL
ReadCache=100
```

Clear the configuration on all controllers:

```
Controller=ALL
ClearConfigurationWithDataLoss=Yes
```

Controller=First

You can use `Controller=First` instead of the `Slot=N` command.

Advantages

- You do not need to determine where controllers are installed.
- You can target the first controller found, a common practice with embedded controllers, because they are always slot 0. (Controllers are sorted by slot number, lowest to highest, and internal controllers before external ones.)

Example

Configure a volume on the embedded controller (or the controller in the lowest slot number, if an embedded controller does not exist):

```
Controller=First
Array=A
Drive=1E:1:1, 1E:1:2
LogicalDrive=1
RAID=1
```

-internal or -external

You can use the command line option `-internal` or `-external` instead of the default option, which is both internal and external controllers.

Advantage

This option filters the discovered controllers, restricting the command to only internal or only external controllers.

Example

When using the `Controller=First` command, if the system had no internal controllers (due to hardware failure or discovery failure) but had an external controller for shared storage, the script would attempt to create a volume on the external controller. To prevent that action, the `-internal` command line argument restricts discovery to only internal controllers:

```
# hpssascripting -i input.txt -internal
Controller=First
Array=A
Drive=1E:1:1, 1E:1:2
LogicalDrive=1
RAID=1
```

Array=Next

You can use `Array=Next` instead of the `Array=A` command.

Advantage

You do not need to know how many arrays exist. The scripting software selects the next valid array identifier.

Example

Add one array to the configuration, regardless of how many arrays already exist:

```
Controller=Slot1
Array=Next
Drive=1E:1:1, 1E:1:2
LogicalDrive=2
RAID=1
```

HP recommends pairing the `Array=Next` command with `LogicalDrive=Next` (on page 13), because you would not create an array without also creating a logical drive, and this additional command does not require you to know how many logical drives already exist.

LogicalDrive=Next

You can use `LogicalDrive=Next` instead of the `LogicalDrive=N` command.

Advantage

You do not need to know how many logical drives exist. The scripting software selects the next valid identifier.

Example

Add one logical drive to Array A, regardless of how many logical drives already exist:

```
Controller=Slot1
Array=A
Drive=1E:1:1, 1E:1:2
LogicalDrive=Next
RAID=1
```

Repeat=N | MAX

You can use this command as part of the logical drive specification.

Advantage

You can quickly divide the available space on an array into multiple volumes without specifying each volume individually.

Example

Create the maximum number of logical volumes the controller supports (each volume is the same size, and the volumes, in total, use all available free space):

```
Controller=Slot1
Array=A
Drive=*
LogicalDrive=Next
Repeat=Max
RAID=1
```

Drive=N

You can use `Drive=N` instead of specifying the actual paths (`Drive=1E:1:1, 1E:1:2`).

Advantage

You can specify the number of drives needed for the configuration, and the scripting software selects the first available drives based on increasing path. This command is most useful when all drives in the configuration are the same size.

Example

Add a logical drive to the configuration using four physical drives:

```
Controller=Slot1
Array=A
Drive=4
LogicalDrive=1
RAID=5
```

Drive=*

You can use `Drive=*` instead of specifying the actual paths (`Drive=1E:1:1, 1E:1:2`).

Advantage

You can use all available drives for the configuration without knowing their number or actual path.

For this command to execute, all configuration rules must be met. For example, if only two physical drives exist, the scripting software uses both drives, but does not create volumes that require more than two drives, such as a RAID 5 volume.

Example

Add a logical drive to the configuration using all unassigned drives:

```
Controller=Slot1
Array=A
Drive=*
LogicalDrive=1
```

RAID=5

In this example, three or more physical drives must be available to support the RAID level.

OnlineSpare=N

You can use OnlineSpare=N instead of specifying the actual paths (OnlineSpare=1E:1:1, 1E:1:2).

Advantage

You can specify the number of spares needed for the configuration, and the scripting software selects the first available drives based on increasing path. This command is most useful when all drives in the configuration are the same size.

Example

Add a logical drive to the configuration using two spare drives:

```
Controller=Slot1
Array=A
Drive=1E:1:1, 1E:1:2
OnlineSpare=2
LogicalDrive=1
RAID=1
```

OnlineSpare=*

You can use OnlineSpare=* instead of specifying the actual paths (OnlineSpare=1E:1:1, 1E:1:2).

Advantage

You can use all available, unassigned drives as spares for the configuration without knowing the number of drives available or their actual paths.

Observe the following guidelines:

- You cannot exceed the controller settings for maximum number of spares allowed in an array, typically eight spares.
- You cannot combine this command with Drive=* (on page 14), because the script processes the Drive specification first and uses all available drives for that action, leaving no unassigned drives for the OnlineSpare request.

Example

Add a logical drive to the configuration using all unassigned drives as spares:

```
Controller=Slot1
Array=A
Drive=1E:1:1, 1E:1:2
OnlineSpare=*
LogicalDrive=1
RAID=1
```

DriveType=SCSI | SAS | SATA | SATASSD | SASSSD

You can specify what type of drive to use in a configuration.

Advantage

You can ensure that only a certain type of drive is used in a configuration where multiple drive types exist.

Example

Create one volume using all existing SAS solid state drives, and create one volume using all existing SATA drives:

```
Controller=Slot1
Array=A
Drive=*
DriveType=SAS SSD
LogicalDrive=1
RAID=0
```

```
Array=B
Drive=*
DriveType=SATA
LogicalDrive=2
RAID=0
```

Size=MAXMBR

You can use this command to limit the size of the volume to 2^{32} blocks (2TiB), which is the largest bootable volume supported on MBR-based systems.

Advantage

You configure the largest possible, bootable volume size regardless of the free space available, provided that the free space is at least 2TiB.

Example

Create an array with a max sized bootable volume from the 3TB disk at 11:1:1:

```
Controller= Slot 1
Array= Next
Drive= 11:1:1
LogicalDrive=Next
RAID= 0
Size= MAXMBR
```

SizeBlocks=N

You can use this option to capture the exact number of 512-byte blocks in a volume, and then to recreate the exact size when playing back a capture.

Advantage

The `Size` value is rounded to the nearest MB. When you capture the same system repeatedly, this rounding can result in the eventual loss of a small amount of volume size. This option eliminates the rounding.

Example

Create an array with volume using exactly 20,971,520 512-byte blocks (exactly 10 GiB) using the disk at 11:1:1:

```
Controller= Slot 1
Array= Next
Drive= 11:1:1
```



```
LogicalDrive=Next  
RAID= 0  
SizeBlocks= 20971520
```

Advanced scripting scenarios

Scenario 6: Clear the configuration on all internal controllers

Task

Clear the current configuration on all internal controllers in the system.

Hardware configuration

Unspecified

Script

```
# hpssascripting -i input.txt -internal

Controller=ALL
ClearConfigurationWithDataLoss=Yes
```

Summary

By combining the advanced commands to specify `-internal` controllers first, and then `ALL` controllers, the user can clear configurations without affecting external controllers attached externally and without having to identify the internal controllers by slot or clear them one at a time.

Scenario 7: Add a simple volume on systems with mixed controller and physical disk configurations

Task

Add a simple, one-disk RAID 0 volume on each system, using one script. However, do not add a new volume to systems booting from an external controller.

Hardware configuration

The administrator has 500 systems, some with embedded controllers, some with controllers in expansion slots, and some booting from an external controller, such as an HP StorageWorks MSA1500cs Modular Smart Array.

Each system has a different physical disk configuration, some disks in an internal drive cage and some in an external storage box. Each system can have a different number of arrays and volumes.

Script

```
# hpssascripting -i input.txt -internal

Controller=First
Array=Next
Drive=1
```

```
LogicalDrive=Next  
RAID=0
```

Summary

By combining the advanced commands to specify `-internal` controllers only, and then `First` controllers, the user does not add volumes to external controllers or to additional internal controllers on the same system. The `Next` command eliminates the need to identify specific arrays and volumes.

Scenario 8: Create multiple arrays on unconfigured systems with mixed physical disk configurations

Task

Add two arrays, each with one volume, to each system. Configure the first array as an OS volume with RAID 1 and a single spare. Configure the second array to use all remaining disks in a RAID 5 volume with no spares.

Hardware configuration

The administrator has 500 unconfigured systems with embedded controllers. Each system has a different physical disk configuration, some disks in an internal drive cage and some in an external storage box.

Script

```
Controller=Slot0  
  
Array=A  
Drive=2  
OnlineSpare=1  
LogicalDrive=1  
RAID=1  
  
Array=B  
Drive=*  
LogicalDrive=2  
RAID=5
```

Summary

The `Slot0` specification targets embedded controllers only. Each array is configured using available drives. Array A uses two available drives, and the script eliminates the need to specify the drives by address. Array B uses all remaining available drives, again without specific drive addresses, but also without the administrator having to specify the number of drives, which can vary for each system.

Scenario 9: Create the maximum size bootable volume

Task

When more than 2TiB of disk space is available, use all possible physical disks to create an array, create the maximum sized boot volume, and then consume the remaining space in another volume.

Hardware configuration

The administrator has an unconfigured system with a controller.

Script

```
Controller= Slot 1
Array= A
Drive= *
LogicalDrive=1    ; consumes the first 2TiB
RAID= 0
Size= MAXMBR
LogicalDrive=2    ; consumes remainder of free space
RAID=0
```

Summary

Without specifying drives by name, the administrator instructs the controller to use available disk space, up to 2TiB, to create the first volume in an array. Any remaining disk space is consumed in a second volume automatically.

Scenario 10: Create a configuration with a specific block size

Task

Identify the size of a logical volume in blocks, and then create a configuration for another logical volume that matches exactly, using block size instead of capacity.

Currently, all Smart Array volumes consist of 512-byte blocks. When you create a custom configuration that uses the `Size` parameter, the script specifies the input size in MB, which rounds the requested value into some number of blocks equal to or slightly greater than the requested size; the exact value depends on the array configuration.

However, if you use the `SizeBlocks` parameter to create the logical volume, then the script specifies the input size in the exact number of blocks, and no rounding occurs. In some cases, the RAID configuration might adjust the number of blocks in the new logical drive.

Hardware configuration

The administrator has a system with a controller and a configured logical volume.

Script output

To identify the size of an existing logical volume, in blocks, observe the output when you capture the existing configuration. Scripting outputs the current size of volumes in block units in a comment.

```
; Logical Drive Specifications
LogicalDrive= 2
RAID= 5
Size= 279964
; SizeBlocks= 573367088
Sectors= 32
StripSize= 512
Caching= Enabled
```

You can then use the size, specified in blocks, to configure another logical volume that is an exact match in size. When using an input script, you can specify either `Size` or `SizeBlocks`, but not both.

Summary

To create a logical drive that uses the exact same configuration of another logical drive, use the `SizeBlocks` parameter to specify the size, thus avoiding the rounding that can occur with the `Size` parameter.

Scenario 11: Configuring HP SmartCache

Task

This script illustrates how to configure HP SmartCaching.

Hardware configuration

The administrator has a system with a controller, physical drives, and logical drives configured.

Script output

```
Action= Configure
Method= Custom

Controller= Slot 1

Array= A
Drive= 2E:1:1, 2E:1:2, 2E:1:3

LogicalDrive= 1
RAID= 5
Size= 139947

CachingArray= B
Drive= 6I:0:5

CachingLogicalDrive= 2
CachedLogicalDrive= 1
RAID= 0
Size= 13993
```

Summary

Caching Logical Drives and Cached data drives must be on the same Controller.

Array A is created with 3 SAS physical drives, containing a single RAID 5 data logical drive (Logical Drive 1). Caching Array B is created using a single SSD drive, containing a single RAID 0 caching logical drive (Logical Drive 2) and is associated with Logical Drive 1.

Acronyms and abbreviations

ACU

Array Configuration Utility

HP SSA

HP Smart Storage Administrator

MBR

master boot record

SAAP

Smart Array Advanced Pack

SAS

serial attached SCSI

SATA

serial ATA

Documentation feedback

HP is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (<mailto:docsfeedback@hp.com>). Include the document title and part number, version number, or the URL when submitting your feedback.